ORBITAL MECHANICS OF PLANETS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 16, 2026

Diagram-based questions on Orbital Mechanics of Planets

Question 1

10.0 points

Examine the diagram below depicting a planet's elliptical orbit around the sun. Using Kepler's second law, describe how the area swept out by a line from the planet to the sun varies over time. Explain how the speed of the planet changes as it moves along different segments of its orbit and why.

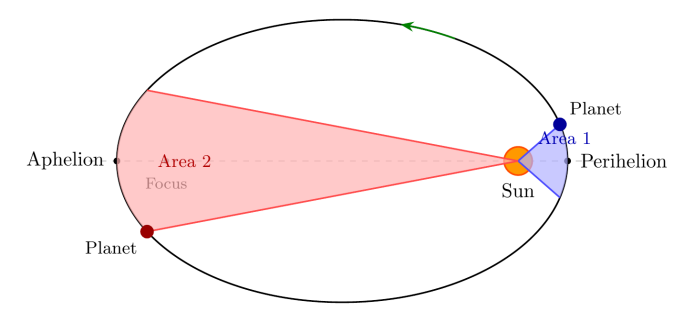

Fig. 1. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: According, Kepler, equal areas; The diagram explicitly labels two sectors with "Area swept in equal time", which directly states the principle of Kepler's second law that the student is asked to explain.; The diagram visually depicts the answer by showing two sectors of approximately equal area, which demonstrates the principle of "equal areas in equal times" that the student is supposed to explain.. A diagram of a planet in an elliptical orbit around the sun. The sun is at one focus of the ellipse. The major axis is drawn as a dashed line, with the 'Perihelion' and 'Aphelion' points labeled at the ends. The other focus is also marked. Show two shaded sectors representing the area swept by the planet's orbit. One sector, labeled 'Area 1', should be near the perihelion. The other sector, labeled 'Area 2', should be near the aphelion. CRITICAL: The two areas must be drawn with clearly and significantly different sizes to avoid visually leaking the 'equal areas in equal times' law. For example, make Area 2 visibly larger than Area 1. The planet should be labeled at the edge of one of the sectors. Do not include any labels that state or imply the areas are equal or are swept in equal times.

Question 2 10.0 points

Examine the diagram below showing a spacecraft executing a Hohmann transfer orbit between two circular orbits around Earth. Calculate the velocity changes (Δv) required at each segment of the transfer. Provide a detailed step-by-step explanation of the calculation.

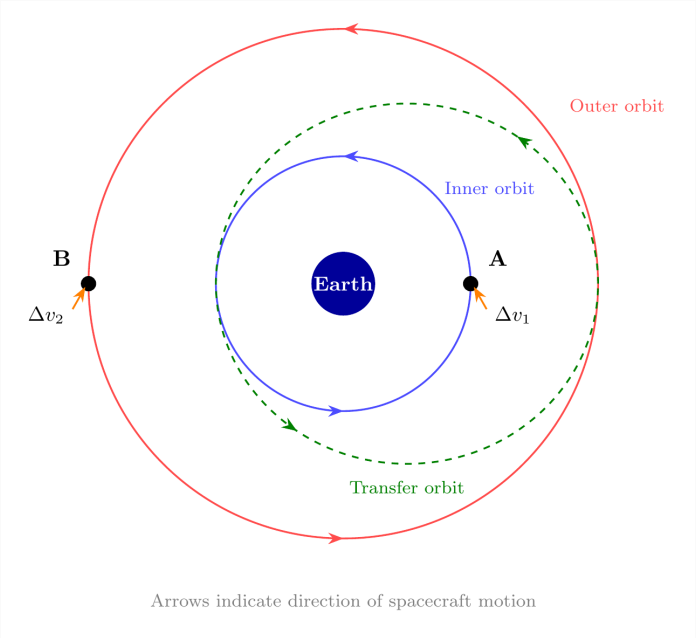

Fig. 2. MANDATORY LABELS — every item below MUST appear as visible text in the diagram: • Δv The diagram will FAIL automated review if any of these are missing. A Hohmann transfer orbit diagram showing two concentric circular orbits around Earth. The inner orbit is the initial circular orbit, and the outer orbit is the final circular orbit. The transfer orbit is an elliptical path that connects the two circular orbits. The diagram includes Earth at the center, the two circular orbits, and the elliptical transfer orbit. Mark the points on the elliptical orbit where the velocity changes (Δv) occur: one at the periapsis (closest point to Earth) and one at the apoapsis (farthest point from Earth). Label these points as Δv1 and Δv2 respectively. Include arrows to indicate the direction of the spacecraft's motion along the orbits. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Step • Examine • Use • Hohmann • Refer • Compute • Final The diagram will FAIL review if any of these appear.

Question 3

10.0 points

Examine the diagram below that illustrates the trajectory of a space probe executing a gravitational slingshot maneuver around a planet. Explain how this maneuver alters the probe's trajectory and velocity, and discuss the conservation laws that apply to this process.

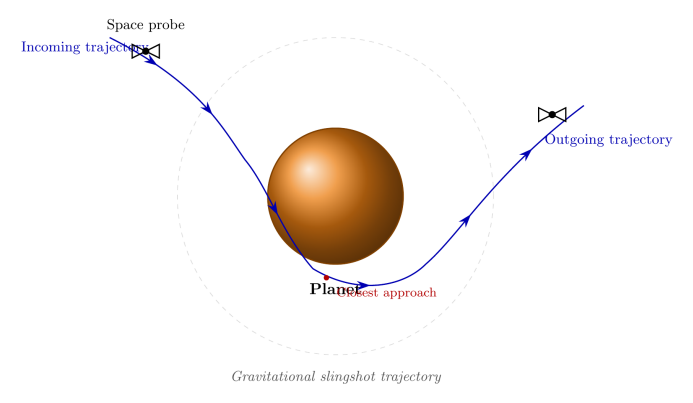

Fig. 3. A diagram showing the trajectory of a space probe performing a

Question 4

10.0 points

Examine the diagram below depicting the five Lagrange points (L1-L5) in the Earth-Sun system. Describe the stability of each point and how satellites can utilize these positions.

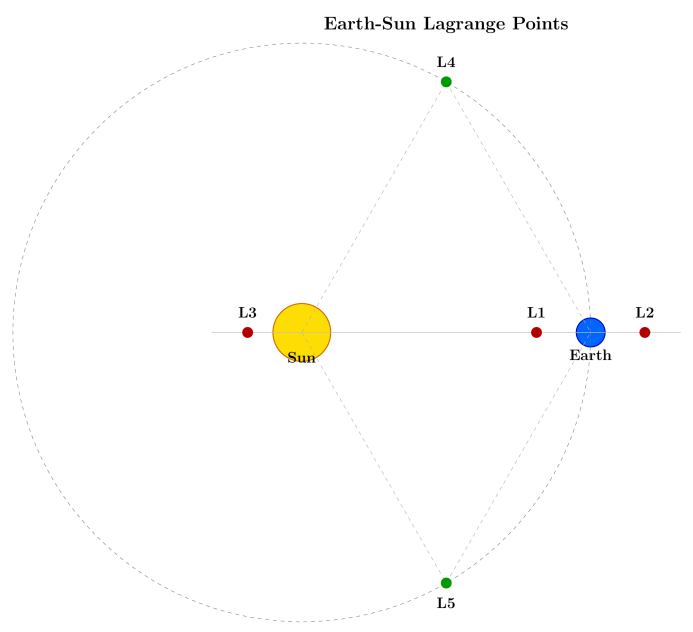

Fig. 4. A diagram

Question 5 10.0 points

Using the diagram below that details Mercury's orbit, explain the phenomenon of orbital precession and its causes, including contributions from both Newtonian mechanics and general relativity, with equations <eq_q_1> and <eq_q_2> resolved.

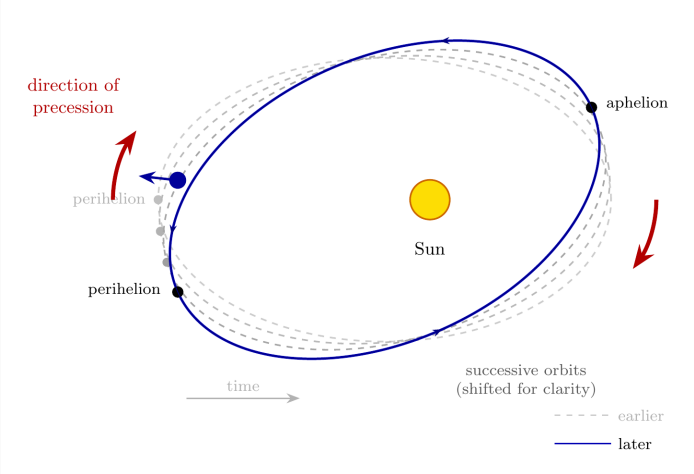

Fig. 5. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: Orbital, Mercury, Newtonian, phenomenon of orbital precession, base editing window; The diagram contains multiple forbidden labels ("Mercury", "Newtonian", "Orbital") and also directly reveals the answer by explicitly labeling the two causes of precession, which the student is required to explain.. Diagram showing an elliptical orbit of a planet around the Sun. The orientation of the ellipse should be shown to precess over time, with successive orbits drawn to illustrate this shift. The perihelion point should be marked for each successive orbit. Key points like the perihelion and aphelion must be labeled. Arrows must indicate the direction of the planet's motion and the direction of the precession. Crucially, there must be annotations that indicate the portion of the precession caused by Newtonian mechanics (e.g., gravitational tugs from other planets) and the portion caused by general relativity. IMPORTANT correction needed: The diagram contains multiple forbidden labels ("Mercury", "Newtonian", "Orbital") and also directly reveals the answer by explicitly labeling the two causes of precession, which the student is required to explain.. Specifically: The diagram contains the forbidden labels "Mercury" (in the title, as. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.